Unit 2 Review

Everything Unit 2 taught, one small example each, to use while you build your Turtle Art Gallery picture.

1. Run a turtle program

Click **Open Graphics** first, then **Run**. Running with no graphics window open gives an error. Every program starts with `import turtle` and ends with `turtle.done()`. Errors show in the terminal, with a line number.

2. Move and turn

forward() moves the turtle that many steps the way it is facing. **left()** and **right()** turn it in place by that many degrees. The way it faces is its **heading**, and it stays set until you turn again.

```
turtle.forward(100)
turtle.left(90)
turtle.forward(100)
```

3. Travel without drawing

(0, 0) is the middle of the window. x goes left and right, y goes up and down. **penup()** lifts the pen, **goto(x, y)** sends the turtle to that spot, and **pendown()** puts the pen back.

```
turtle.penup()
turtle.goto(-100, 50)
turtle.pendown()
turtle.forward(80)
```

4. Color and pen size

Set these before you draw. They stay set until you change them.

```
turtle.color("purple")
turtle.pensize(5)
turtle.forward(60)
```

5. For loops

A for loop repeats its **loop body**, the indented lines under it. **range(n)** sets how many times. A line that isn't indented runs once, after the loop.

```
for i in range(5):
    turtle.forward(20)
    turtle.penup()
    turtle.forward(20)
    turtle.pendown()
```

6. Regular shapes

The turns add up to one full circle, so the turn is **360 / sides**.

```
sides = 6
for i in range(sides):
    turtle.forward(50)
    turtle.left(360 / sides)
```

7. The loop variable

The loop variable (here `i`) holds the pass number, starting at 0. You can use it in math, so each pass is different.

```
for i in range(4):
    print(i)
    turtle.forward(10 + i * 20)
    turtle.left(90)
```

8. Nested loops

A nested loop is a loop inside a loop. The **inner loop** draws one copy. The **outer loop** repeats it. The move between copies lines up with the inner `for`, so it runs once per copy. Total repeats = outer × inner.

```
for square in range(5):
    for side in range(4):
        turtle.forward(40)
        turtle.left(90)
    turtle.penup()
    turtle.forward(60)
    turtle.pendown()
```

9. While loops

`while True:` repeats forever. Stop it with **Ctrl + C** in the terminal. A **while** loop with a condition stops on its own.

```
count = 0
while count < 3:
    turtle.forward(40)
    turtle.right(120)
    count = count + 1
```